2023-24

S5 MATH EP

M1

Name: _____

Class: _____( )

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Five First term Examination

MATHEMATICS Extended Part
Module 1 (Calculus and Statistics)
Question-Answer Book

10/01/2024

8:15 am – 10:45 pm (2½ hours)

This paper must be answered in English

	Marker Use Only
Question	Mark
1	
2	
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INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. The paper consists of Two sections A and B.

3. Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string **INSIDE** this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or given to 4 decimal places.

7. No extra time will be given to candidates for writing your Name, Class and Class number or filling in the question number boxes after 'Time is up' announcement.

Total			

Section A (50 marks)

1. (a) Expand e^{-2x} in ascending powers of *x* as far as the term in x^3.

(b) Hence, or otherwise, find the coefficient of x in the expansion of $(1-\frac{1}{x}+\frac{1}{x^2})e^{-2x}$. (6 marks)

2. Consider the curve C : $y = x\sqrt{4x+1}$, where $x > -\frac{1}{4}$.

(a) Find $\frac{dy}{dx}$.

(b) Find the number of tangents to C which is/are perpendicular to the straight line $4x + y = 0$. (8 marks)

3. The rate of change of the volume V (in cm^3) of a certain substance can be modeled by

$$\frac{dV}{dt} = \frac{3(60-t)}{e^{\frac{t}{60}}+t} \qquad (t \geq 0)$$

where t is the time measured in seconds. It is known that $V = 30$ when $t = 0$.

(a) Let $u = 1 + te^{-\frac{t}{60}}$. Find $\frac{du}{dt}$.

(b) Hence, or otherwise, express V in terms of t. (8 marks)

4. Let A and B be two events with $P(A \mid B) = \frac{1}{2}$, $P(B \mid A) = \frac{3}{7}$, and $P(B) = k$.

(a) Express $P(A \cap B)$ in terms of k.

(b) If $P(A \cup B) = 0.85$, find the value of k. (6 marks)

5. Suppose A and B are two events. Let A' and B' be the complementary events of A and B respectively. $P(A \mid B) = 0.8$, $P(A \cap B') = 0.58$ and $P(A \cap B) = 0.22$.

(a) Find $P(A)$, $P(B)$ and $P(A \cup B)$.

(b) Are A and B' are independent? Explain your answer. (7 marks)

6. The table below shows the probability distribution of a discrete random variable X, where a and b are constants.

X	1	3	5	8	10
$P(X=x)$	a	0.15	0.1	0.4	b

It is given that E(X) = 4.95.

(a) Find the values of a and b.

(b) Find $E((3 – 2X)^2)$ and Var (3 – 2X). (7 marks)

7. The table below shows the probability distribution of a discrete random variable X, where m and k are constants and m is an integer.

X	– 1	2	m	6
$P(X=x)$	0.1	k	0.4	0.2

(a) Find the value of k. Hence express Var (X) in terms of m.

(b) It is given that Var(X) = 3.69. Find Var($8X^2 - 7$). (8 marks)

Section B (50 marks)

8. There are 5 white balls and 7 black balls in a bag. In a game, two players A and B, starting from A, take turns to draw a ball from the bag at random without replacement. A wins if A draws a white ball before B draws a black ball. On the other hand, B wins if he draws a black ball before A draws a white ball.

   (a) Find the probability that B will win

   (i) on his 1st draw;

   (ii) on his 2nd draw;

   (iii) on his 3rd draw.

   (b) Find the probability that A will win within 2 draws. (8 marks)

9. A man travels to work by bus, car or on foot with probabilities $\frac{1}{6}$, $\frac{1}{3}$ and $\frac{1}{2}$ respectively. For each type of transport his probabilities of being late are $\frac{1}{5}$, $\frac{1}{4}$ and $\frac{1}{20}$ respectively.

(a) Find the probability that he was late on a particular morning.

(b) If he arrives late on that morning, find the probability that he came by bus. (7 marks)

10. There is a blood test for a certain disease E. If a person having the disease E is tested, the test has a probability 7% of showing a negative result. If a person not having the disease E is tested, the test has a probability 3% showing a positive result. It is estimated that 4% in a town have the disease E.

(a) If every person in the town is tested, find the proportion of people with positive results in the blood test.

(b) If a person gets a positive result in the blood test, find the probability that he / she actually has the disease E.

(c) If a person gets a negative result in the blood test, find the probability that he / she actually does not have the disease E.

(8 marks)

11. In an experiment, rate of change of the amount of oxygen produce (in cm^3) by machine A and B are respectively as follows:

$$f(t) = \ln(e^{2t} - t) \text{ and } g(t) = \frac{255-15t^2}{7(1+t)},$$

where t ($2 \leq t \leq 4$) is the time measured in minutes.

(a) Using the trapezoidal rule with 4 sub-intervals, estimate the total amount of oxygen produced by Machine A from t = 2 to t = 4. (3 marks)

(b) Determine whether the estimate in (a) is an over-estimate or under-estimate. Explain your answer. (4 marks)

(c) Find $\int_2^4 \frac{17-t^2}{1+t} dt$. (3 marks)

(d) Ryan claims that the total amount of oxygen produced by machine A from t = 2 to t = 4 is more than that of machine B. Do you agree? Explain your answer. (3 marks)

12. The average value of flats \$V (in ten thousand) in a certain city in the first six months of a certain year can be modelled by

$$V = \frac{800}{1+\alpha t e^{1+\beta t}} \ ,$$

Where t $(0 \leq t \leq 6)$ is the number of month elapsed since the start of the year and α and β are constants.

(a) Express $\ln\left(\frac{800}{Vt} - \frac{1}{t}\right)$ as a linear function of t. (3 marks)

(b) It is given that the slope and the intercept on the vertical axis of the graph of the linear function obtained in (a) are – 0.2 and $\ln\frac{e}{200}$ respectively.

(i) Find α and β.

(ii) Will the average value of flats be less than 780 ten thousand during the first 6 months in the year? Explain your answer.

(iii) Describe how the rate of change of the average value of flats in the city varies during the first 6 months in the year. Explain your answer. (11 marks)

---End of Paper---